

TOUR DE MATHS

LEG 6



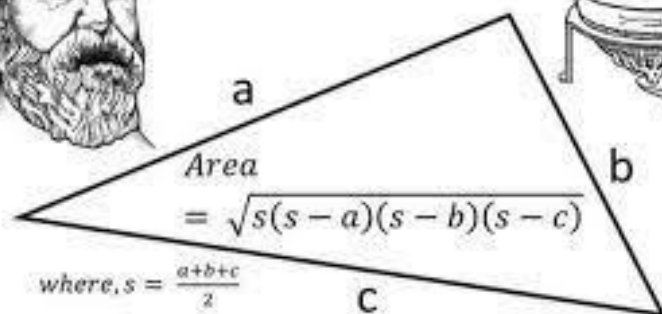
ST DAVIDS MARIST INANDA

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CASIO



Heron's Formula



Questions

1. What number when divided by 6, 11 and 13 will have a remainder of $\frac{3}{2}$?

2. Evaluate: $(1 - 2 + 3 - 4 + 5 - \dots - 100) \left(\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} \times \dots \times \frac{99}{100} \right)$

3. Two numbers **A** and **B** have the following in common:

$$\text{HCF} = 2 \times 5^3 \text{ and } \text{LCM} = 2^2 \times 5^4 \times 7.$$

The prime factors for **A** and **B** are: $A = 5^a \times b^2$ and $B = 2 \times 5^4 \times c$

Write down the numerical value of $a \times b \times c$.

*Hint: **HCF = Highest common factor**

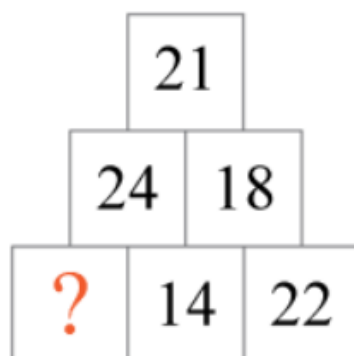
LCM = Lowest common multiple

4. If $\frac{x}{y} = \frac{6}{5}$ Calculate the value of $\frac{x^2 + y^2}{x^2 - y^2}$

5. If $A - B = m$ and $n + A = B + C$

Write $C + 3B - 3A$ in terms of m and n .

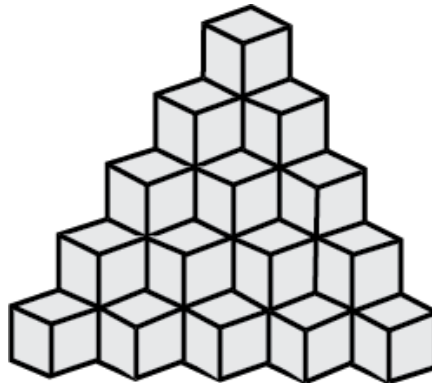
6. Find the missing number.



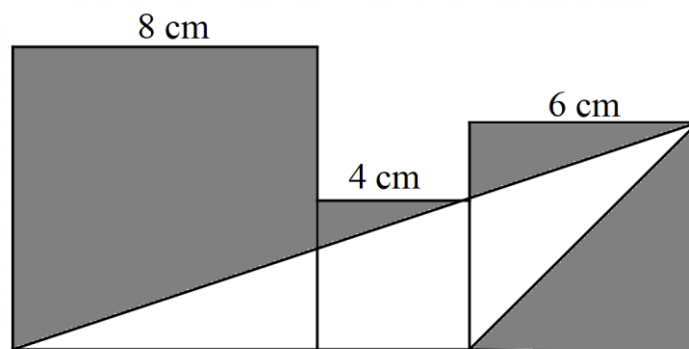
7. A grandfather, two fathers, two sons and one grandchild went together to buy movie tickets. Everyone bought one ticket for themselves. What is the least number of tickets that could be bought?

Hint: All these people are from the same family

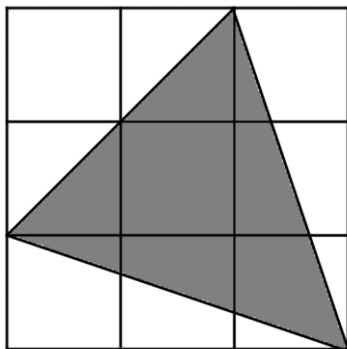
8. The model of a tower shown below is made up of unit cubes stacked on top of each other. How many unit cubes in all are used to form the tower?



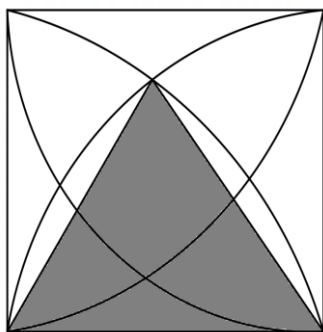
9. The figure below is made up of three squares with sides 8 cm, 4 cm, and 6 cm. Calculate the area of the shaded region



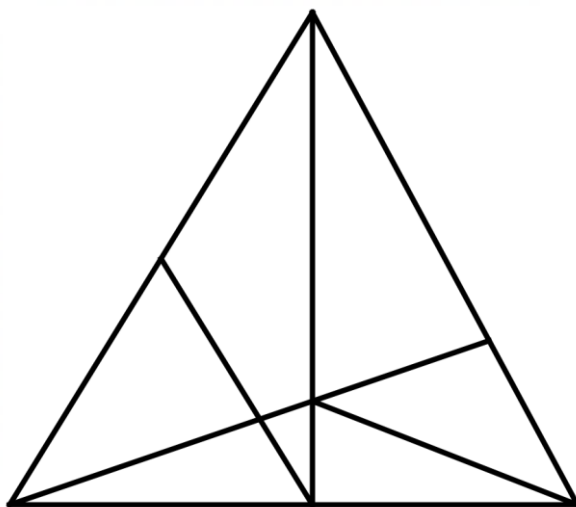
10. Find the ratio of the shaded region to the unshaded region. Each block has a side of 1 unit.



11. Each arc is a quarter of a circle with its centre at a vertex (corner) of the square. Given the square has side length of 1 unit. Find the area of the shaded triangle. **(Leave your answer as a fraction)**

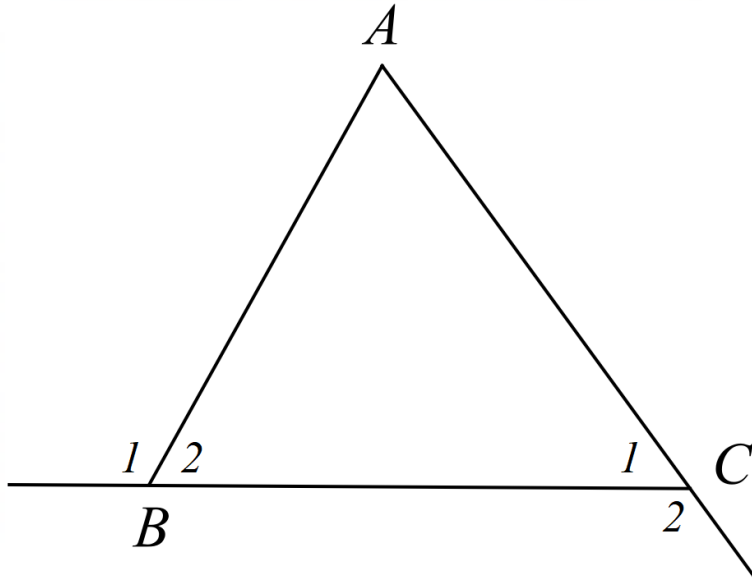


12. How many triangles can you count below in this picture?



13. In the triangle below $\hat{B}_1 = x$ and $\hat{C}_2 = y$. Find the numerical value of $x + y$

$$\hat{A} + \hat{B}_1 + \hat{C}_2 = 260^\circ$$



14. The symbol '!', an exclamation mark, is known as a factorial in Mathematics.

Examples:

$$8! = 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$$

$$7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$$

$$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1$$

For any given factorial, you multiply all the consecutive natural numbers down to 1.

Calculate $\frac{200!}{197! \times 3!}$ without a calculator

15. The numbers 49, 29, 9, 40, 22, 15, 53, 33, 13 and 47 are grouped in the pairs so that the sum of each pair is the same. Which number will be paired with 15?

16. Study the **consecutive** fractions below and answer the following question.

Fraction: 1 $\frac{62}{43} = 1 + \frac{19}{43} = 1 + \frac{1}{2 + \frac{1}{3 + \frac{4}{5}}}$	Fraction: 2 $\frac{75}{52} = 1 + \frac{23}{52} = 1 + \frac{1}{2 + \frac{1}{3 + \frac{5}{6}}}$
Fraction: 3 $\frac{88}{61} = 1 + \frac{27}{61} = 1 + \frac{1}{2 + \frac{1}{3 + \frac{6}{7}}}$	

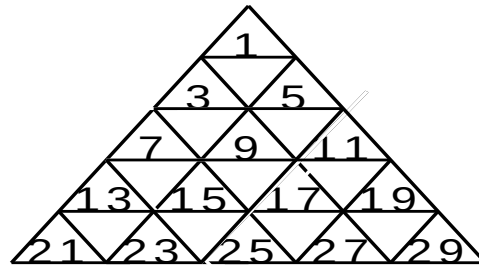
Calculate the sum of a , b , and c for the fraction below. a, b and c are positive integers

$$\frac{a}{898} = 1 + \frac{1}{2 + \frac{1}{3 + \frac{b}{c}}}$$

17. If $b^c = 81$ $b^a = 9$ $c^a = 5$ Calculate the numerical value of C^c
 a, b and c are positive integers

18. The number 23 is vertically below 7 in row 5.

Write down the number that will be exactly below 7 in row number 201.



19. Jimmy wants to create a three-digit number lock code for his security company using the digits.

0; 1; 2; 3; 4 and 5.

How many even-number codes can Jimmy create if the digits are not repeated?



20. Tom and Peter can finish work together in 12 hours. Peter and Sam can finish their work in 8 hours. Tom and Sam can finish their work in 6 hours. How long (in hours) will it take Peter if he works all by himself?

*****END*****

FUN FACT

Rene Descartes

The idea of graphing with coordinate axes dates all the way back to the 1600s, gets the credit for coming up with the two-axis system we use today.

The story goes that he lay in bed and watched flies crawling over tiles on the ceiling. He realized that he could describe a fly's position using the intersecting lines of the tiles. The system is often called the "Cartesian coordinate system" in his honor.

